ECON 5021 Unit 2: SAQ Flashcards (Short)

Bare minimum per SAQ: memorize the **hook**, then the 1–3 lines under it. Longer versions are in `ECON5021_Unit2_SAQ_Flashcards.pdf`.

1. L8 Q1: μ minimizes $\mathrm{E}(y-a)^2$

Hook: add and subtract μ; the cross term is 0.

$\mathrm{E}(y-a)^2 = \mathrm{E}(y-\mu)^2 + (\mu-a)^2$ (cross term $2(\mu-a)\mathrm{E}(y-\mu)=0$). Minimized at $a=\mu$. ■

2. L8 Q2: CEF is the best predictor

Hook: (a) definition, (b) LIE, (c) Card 1 trick with $h = m - g$.

(a) $\mathrm{E}(u|x) = m - m = 0$. (b) $\mathrm{E}(hu) = \mathrm{E}[h\,\mathrm{E}(u|x)] = 0$.
(c) $\mathrm{E}(y-g)^2 = \mathrm{E}u^2 + 2\mathrm{E}[u(m-g)] + \mathrm{E}(m-g)^2 = \mathrm{E}u^2 + \mathrm{E}(m-g)^2$, minimized at $g=m$. ■

3. L8 Q3: residual facts

Hook: FOC is $\sum x_{ik}e_i = 0$; then $k=1$; average; Cov.

(a) $k=1$: $\sum e_i = 0$. (b) $\bar{e} = \bar{y} - \bar{x}'b = 0$. (c) $\widehat{\mathrm{Cov}}(x_k, e) = \frac{1}{n}\sum x_{ik}e_i - \bar{x}_k\bar{e} = 0 - 0$. ■

4. L8 Q4: bivariate $b_2 = \mathrm{Cov}/\mathrm{Var}$

Hook: factor out n; the n's cancel; 2×2 inverse; row 2.

(a) $X'X = n\begin{bmatrix}1 & \bar{e}\\ \bar{e} & \overline{ee}\end{bmatrix}$, $X'Y = n\begin{bmatrix}\bar{w}\\ \overline{ew}\end{bmatrix}$, so the n's cancel in $(X'X)^{-1}X'Y$.

(b) Inverse $= \frac{1}{\overline{ee}-\bar{e}^2}\begin{bmatrix}\overline{ee} & -\bar{e}\\ -\bar{e} & 1\end{bmatrix}$; row 2: $b_2 = \dfrac{\overline{ew}-\bar{e}\bar{w}}{\overline{ee}-\bar{e}^2} = \dfrac{\widehat{\mathrm{Cov}}(e,w)}{\widehat{\mathrm{Var}}(e)}$. ■

5. L9 Q1: OLS is unbiased

Hook: substitute Y; A3; LIE.

$b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon} \Rightarrow \mathrm{E}(b|X) = \beta + (X'X)^{-1}X'\underbrace{\mathrm{E}(\boldsymbol{\varepsilon}|X)}_{0} = \beta$; $\mathrm{E}b = \mathrm{E}[\mathrm{E}(b|X)] = \beta$. ■

6. L9 Q2: $\mathrm{Var}(b|X) = \sigma^2(X'X)^{-1}$

Hook: $b-\beta = A\boldsymbol{\varepsilon}$; $A\,\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}')A'$; $\sigma^2 I$; cancel.

(a) $b = \beta + (X'X)^{-1}X'\boldsymbol{\varepsilon}$. (b) $\mathrm{Var}(b|X) = (X'X)^{-1}X'\mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|X)X(X'X)^{-1}$.
(c) $\mathrm{Var}(\boldsymbol{\varepsilon}|X) = \mathrm{E}(\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}'|X) - 0$ since $\mathrm{E}(\boldsymbol{\varepsilon}|X) = 0$.
(d) Homoskedastic (diagonal σ^2) + no autocorrelation (off-diagonal 0) $= \sigma^2 I$, so $\sigma^2(X'X)^{-1}X'X(X'X)^{-1} = \sigma^2(X'X)^{-1}$. ■

7. L9 Q3: LLN and CLT

Hook: $X'\boldsymbol{\varepsilon} = n\bar{w}$; LLN $\Rightarrow$ consistent; CLT $\Rightarrow$ normal.

(a) $b = \beta + (X'X/n)^{-1}\bar{w}$. (b) LLN: $\bar{w}\to 0$, so $b\to\beta$ (consistent).
(c) CLT: $\bar{w}$ is approximately normal, so $b \overset{a}{\sim} N(\beta, \sigma^2(X'X)^{-1})$ even without normal errors.

8. L10 Q1: $P_X H = 0$ and $a = 0$

Hook: $Xb - Xb$; then $b - b$.

(a) $P_X = X(X'X)^{-1}X'$. (b) $P_X(Y - Xb) = Xb - Xb = 0$.
(c) Residuals are orthogonal to every column of X. (d) $a = (X'X)^{-1}X'(Y-Xb) = b - b = 0$. ■

9. L10 Q2: projecting on a column of ones

Hook: every entry $1/n$, then $\bar{x}$, then $x_i - \bar{x}$.

(a) P_Z: $n\times n$, every entry $1/n$. (b) $P_Z X$: $n\times 1$, every entry $\bar{x}$. (c) $X^* = M_Z X$: $n\times 1$, entry i is $x_i - \bar{x}$.

10. L11 Q1: standardized estimate

Hook: $N(0,1)$ with σ; t_{n-K} with s.

$\dfrac{b_k-\beta_k}{\sqrt{\sigma^2[(X'X)^{-1}]_{kk}}} \sim N(0,1)$; replacing σ^2 with $s^2=\dfrac{e'e}{n-K}$ gives $\dfrac{b_k-\beta_k}{\widehat{SE}(b_k)} \sim t_{n-K}$.

11. L11 Q2: s^2 is unbiased

Hook: $M'=M$, $MM=M$, $\text{tr}M=n-K$.

(a) $e'e=\boldsymbol{\varepsilon}'M'M\boldsymbol{\varepsilon}=\boldsymbol{\varepsilon}'M\boldsymbol{\varepsilon}$ (symmetric: $M'=M$; idempotent: $MM=I-2P+P=M$).
(b) $\text{E}(e'e|X)=\text{tr}(M\sigma^2 I)=\sigma^2[n-\text{tr}((X'X)^{-1}X'X)]=\sigma^2(n-K)$, so divide by $n-K$. ■

12. L11 Q3: bivariate $\text{Var}(b_2)$

Hook: $(2,2)$ entry of $(X'X)^{-1}$; the determinant is $n\,SST_x$.

$\text{Var}(b_2|X)=\dfrac{\sigma^2 n}{n\sum x_i^2-(\sum x_i)^2}=\dfrac{\sigma^2}{\sum(x_i-\bar{x})^2}$. Factors: $\sigma^2\uparrow$ worse; $n\uparrow$ better; spread in $x\uparrow$ better.

13. L11 Q4: multiple $\text{Var}(b_3)$

Hook: σ^2 over $(1-R_3^2)\,SST_3$.

$\text{Var}(b_3|X)=\dfrac{\sigma^2}{(1-R_3^2)SST_3}$. More precise with a smaller σ^2 (noise), a bigger SST_3 (variation, n), or a smaller R_3^2 (less correlation with the other x's).

14. L11 Q5: multicollinearity

Hook: $R_k^2\uparrow\Rightarrow$ Var $\uparrow$; dropping the variable $\Rightarrow$ OVB.

(a) High R_k^2 makes $1-R_k^2\to 0$, so the SE grows (still unbiased).
(b) Dropping x_j lowers the variance, but x_j goes into ε, so $\text{E}(\varepsilon|X)\neq 0$: omitted variable bias. You trade bias for variance.

15. L12 Q1: log-level, $\log fare = 4.666+0.043\,dist$

Hook: exponentiate; $\beta\cdot fare$; 1 mile $=0.01$.

$fare=e^{4.666+0.043dist}$; $\partial fare/\partial dist=0.043\,fare$; $100\frac{\partial fare}{fare}=4.3(0.01)=0.043\%$. ■

16. L12 Q2: level-log, $fare = 41+66\log dist$

Hook: β/x; $\beta/100$ per 1%; units only change the intercept.

$\partial fare/\partial dist=66/dist$; $\partial fare=\frac{66}{100}(\%\Delta dist)=\$0.66$ per 1%.
Miles: $41+66\log(m/100)=(41-66\log 100)+66\log m$, so the slope is unchanged. ■

17. L12 Q3: log-log, $\log fare = 4.25+0.40\log dist$

Hook: elasticity: $\%\Delta y=\beta\,\%\Delta x$.

$fare=e^{4.25}dist^{0.40}$; $\partial fare/\partial dist=0.40\,fare/dist$; $\%\Delta fare=0.40\times 10=4\%$. ■